Per-Set Bias Analysis

Decomposing synthetic positivity bias by SSR reference set.

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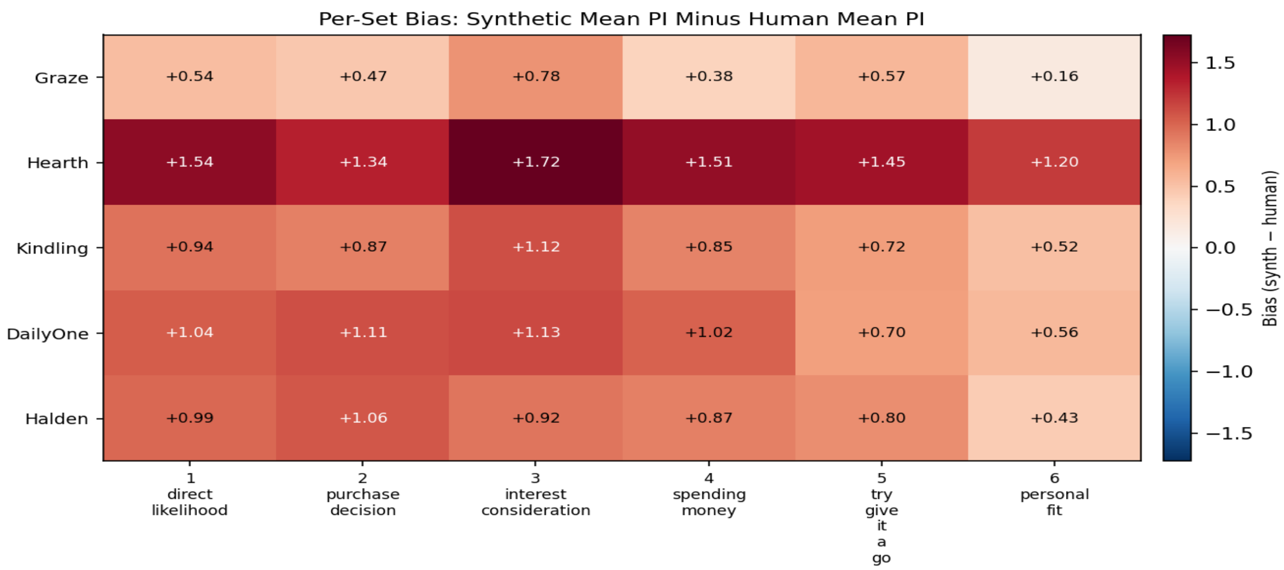
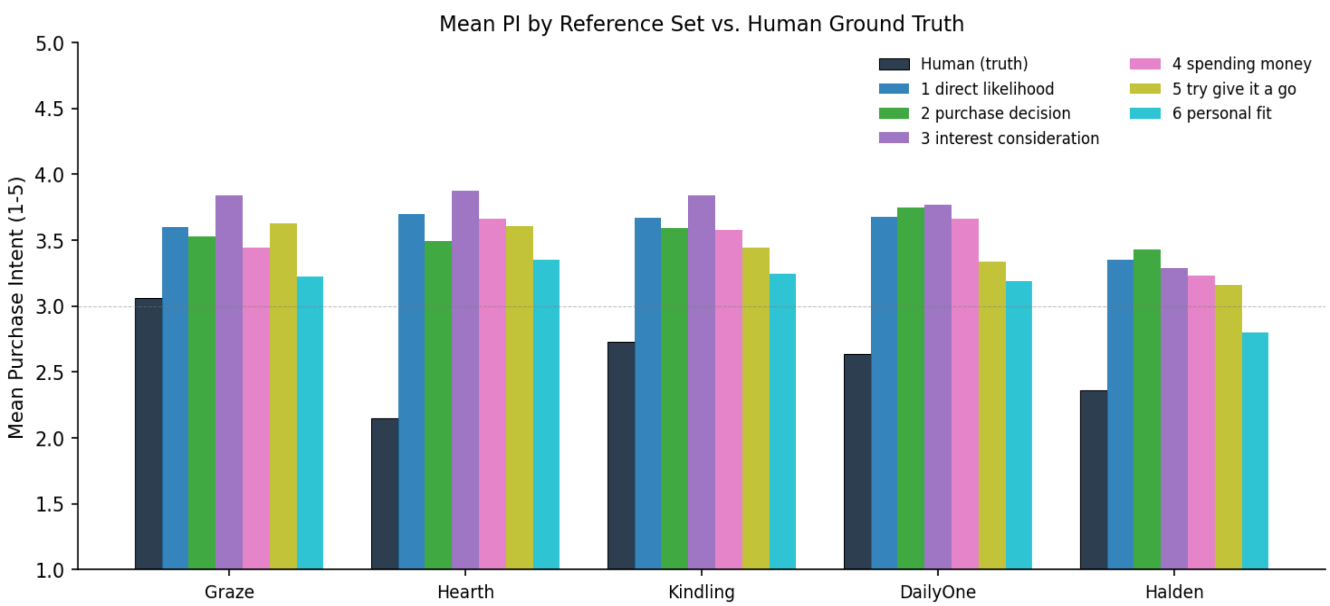
What this report does. The pipeline averages PMFs across 6 reference sets to produce its final Likert distribution. This analysis breaks that average apart and shows what each set *individually* would have produced. Sets that overshoot human ground truth more than others are candidates to drop or down-weight.

Bias matrix

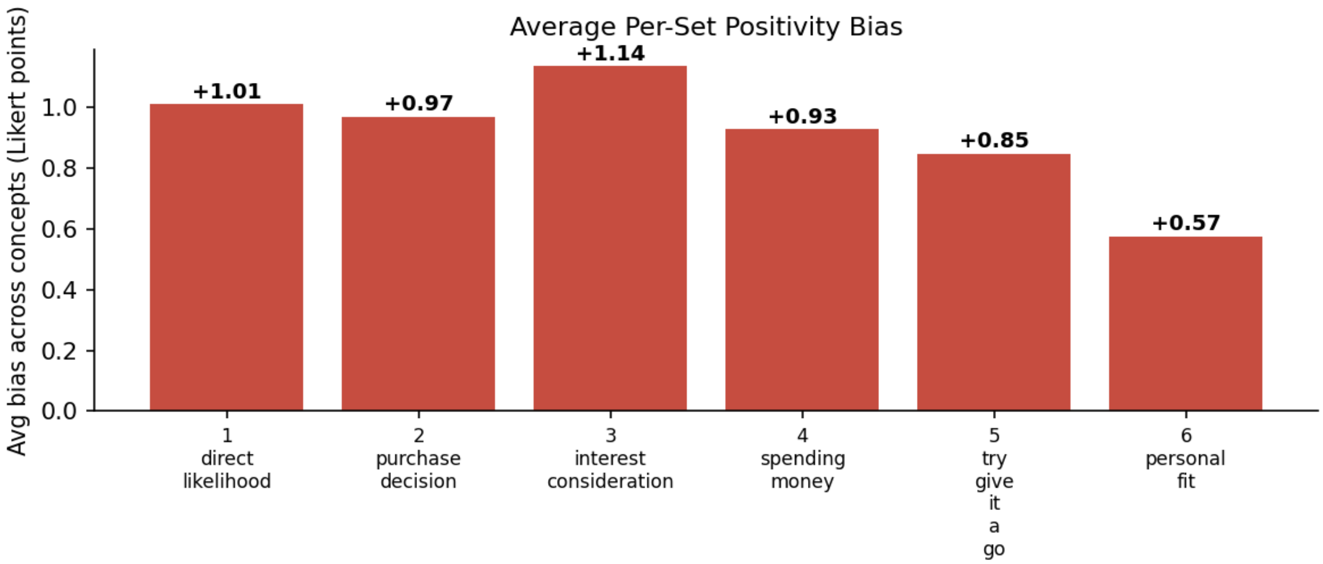


Cells show **synthetic_set_mean** – **human_mean** for each (concept, set) pair. Red = synthetic overshoots human; blue = synthetic undershoots. Larger absolute values = stronger bias.

Per-set means vs. human ground truth



Average bias by reference set



Reference Set	Avg Bias (Likert pts)	Verdict
3_interest_consideration	+1.14	Worst overshooter
1_direct_likelihood	+1.01	
2_purchase_decision	+0.97	
4_spending_money	+0.93	
5_try_give_it_a_go	+0.85	
6_personal_fit	+0.57	Mildest

Drop-one simulation

What happens to mean bias and ranking if we drop a single reference set and average the remaining 5?

Dropped Set	New Avg Bias	∆ vs Baseline	Ranking Preserved?
(none — baseline)	+0.91	—	✗ (Hearth > DailyOne > Kindling > Graze > Halden
3_interest_consideration	+0.87	-0.04	✗ (Hearth > DailyOne > Kindling > Graze > Halden
1_direct_likelihood	+0.89	-0.02	✗ (Hearth > DailyOne > Kindling > Graze > Halden
2_purchase_decision	+0.90	-0.01	✗ (Hearth > Kindling > Graze > DailyOne > Halden
4_spending_money	+0.91	-0.00	✗ (Hearth > Graze > Kindling > DailyOne > Halden
5_try_give_it_a_go	+0.92	+0.01	✗ (Hearth > DailyOne > Kindling > Graze > Halden
6_personal_fit	+0.98	+0.07	✗ (Hearth > DailyOne > Kindling > Graze > Halden

Per-concept detail

Graze

Reference Set	Mean PI	Bias vs Human
Human (truth)	**3.06**	—
1_direct_likelihood	3.60	+0.54
2_purchase_decision	3.53	+0.47
3_interest_consideration	3.84	+0.78
4_spending_money	3.44	+0.38
5_try_give_it_a_go	3.63	+0.57
6_personal_fit	3.23	+0.16
Avg of all 6 sets	**3.54**	**+0.48**

Hearth

Reference Set	Mean PI	Bias vs Human
Human (truth)	**2.15**	—
1_direct_likelihood	3.70	+1.54
2_purchase_decision	3.49	+1.34
3_interest_consideration	3.88	+1.72
4_spending_money	3.66	+1.51
5_try_give_it_a_go	3.60	+1.45
6_personal_fit	3.35	+1.20
Avg of all 6 sets	**3.61**	**+1.46**

Kindling

Reference Set	Mean PI	Bias vs Human
Human (truth)	**2.73**	—
1_direct_likelihood	3.67	+0.94
2_purchase_decision	3.59	+0.87
3_interest_consideration	3.84	+1.12
4_spending_money	3.58	+0.85
5_try_give_it_a_go	3.45	+0.72
6_personal_fit	3.24	+0.52
Avg of all 6 sets	**3.56**	**+0.84**

DailyOne

Reference Set	Mean PI	Bias vs Human
Human (truth)	**2.64**	**—**
1_direct_likelihood	3.68	+1.04
2_purchase_decision	3.75	+1.11
3_interest_consideration	3.77	+1.13
4_spending_money	3.66	+1.02
5_try_give_it_a_go	3.34	+0.70
6_personal_fit	3.19	+0.56
Avg of all 6 sets	**3.56**	**+0.93**

Halden

Reference Set	Mean PI	Bias vs Human
Human (truth)	**2.36**	**—**
1_direct_likelihood	3.35	+0.99
2_purchase_decision	3.43	+1.06
3_interest_consideration	3.29	+0.92
4_spending_money	3.23	+0.87
5_try_give_it_a_go	3.16	+0.80
6_personal_fit	2.80	+0.43
Avg of all 6 sets	**3.21**	**+0.85**

Recommendations

Large spread across sets (0.56 Likert pts). **3_interest_consideration** is a clear outlier (+1.14 bias). Try dropping it and re-running the average. This may meaningfully reduce positivity bias without requiring a full pipeline re-run.

What this analysis can't tell you. Per-set bias is computed against this single 5-concept pilot. Set-level fixes that work here may not generalize to other categories or demographics. Don't permanently drop sets based on one validation run — instead, treat this as a hypothesis to test against your next validation study.